Computational Biology

Lecture 4: *Sequence Re-assembly*

```
GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG
GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG
GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG
GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG
GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG

GGTTAAA   CTCTACTCGGCATG

GAGCA   GGTTA   CTACT   GTTAA   CATGT   TCGGC
TACTC   TTAAA   CCGTC   GGTTA   GGTTA   ACTCG
GGCAT   CATGT   CTACT   GTTAA   TTAAA   TGTCC
GGCAT   GAGCA   TCTCG   AGCAA   GCATG   CTCTA
AGCAA   CGTCT   TCCCG   CTCGA   CGAGC   CAACG
```

Digital trees, tries, suffix trees

Outline

1. **Reconstruction**
2. Digital Trees
3. Suffix Trees
4. Repeats

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GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG
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GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG

GGTTAAA   CTCTACTCGGCATGT

GAGCA   GGTTA   CTACT   GTTAA   CATGT   TCGGC
TACTC   TTAAA   CCGTC   GGTTA   GGTTA   ACTCG
GGCAT   CATGT   CTACT   GTTAA   TTAAA   TGTCC
GGCAT   GAGCA   TCTCG   AGCAA   GCATG   CTCTA
AGCAA   CGTCT   TCCCG   CTCGA   CGAGC   CAACG
```


DNA Sequencing

- In modern whole shotgun genome sequencing the full genome is broken into small pieces
- The pieces are then read by a sequencing machine
- This reads short sections (100–1000) bases
- The **reads** are then assembled to construct the full genome

Sequencing and Assembly

```
GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG
GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG
GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG
GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG
GGTTAAATCCTCTACTCGGCATGTCCCGTCTCGAGCAACG

GGTTAAA   CTCTACTCGGCATGTCCCGTCTCGAGCAACG

GAGCA   GGTTA   CTACT   GTTAA   CATGT   TCGGC
TACTC   TTAAA   CCGTC   GGTTA   GGTTA   ACTCG
GGCAT   CATGT   CTACT   GTTAA   TTAAA   TGTCC
GGCAT   GAGCA   TCTCG   AGCAA   GCATG   CTCTA
AGCAA   CGTCT   TCCCG   CTCGA   CGAGC   CAACG
```

Reads and Contigs

- In the language of DNA reconstruction we are given a set of **reads**
- Our job is to reconstruct them into **contigs**
- The problem is complicated because there are errors in the reads (0.1-2%)
- Shotgun was important in speeding up the human genome project (there is some argument about how important)
- What data structure should we use?

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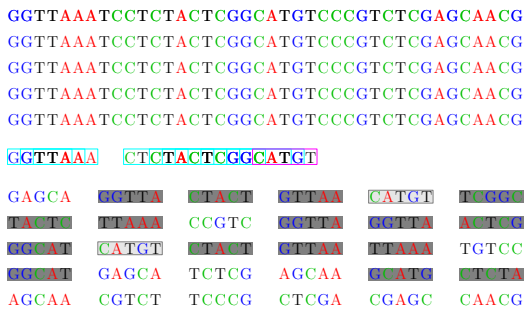
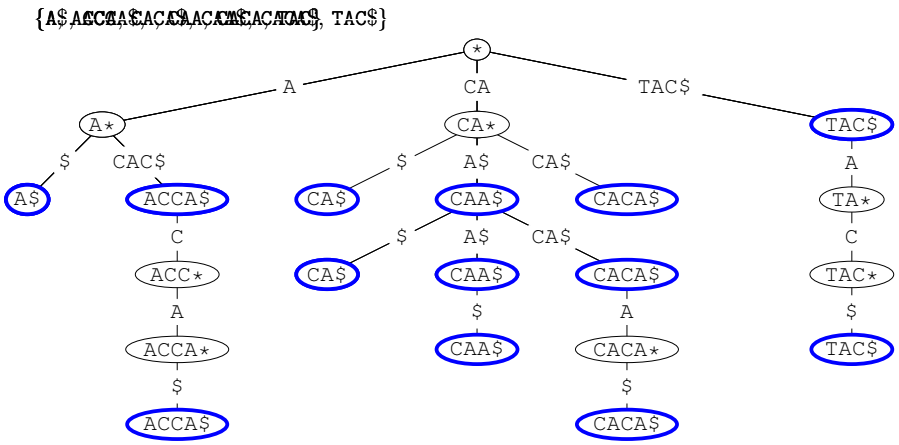


Digital Trees or Tries

- A **Trie** (pron. 'try') or **digital tree** is a multiway tree often used for storing large sets of words
- They are trees with a possible branch for every letter of an alphabet
- Their names comes from the word *re**trie**val*
- Tries usually compactify the edges in the tree
- All words end with a special letter "$"

Trie



General Uses of Tries

- Tries are yet another way of implementing sets
- They provide quick insertion, deletion and find
- Typically considerably quicker than binary trees and hash tables
- They are particularly good for spell checkers, completion algorithms, longest-prefix matching, hyphenation
- Each search finds the longest match between the words in the set and the query

Tries for Sequence Analysis

- Tries can be used to store DNA or proteins
- For reconstruction of DNA tries are very useful for finding the next fitting section
- They are useful when there are errors, as by searching a small sub-tree you can find a good match (same reason they are good for completion and spell checking)
- For DNA you have a four way (5 way?) tree, which means your tree is deep but it doesn't waste so much memory

Trie for 31 Most Common English Words

	0	1	2	3	4	5	6	7	8	9	10
$		I$				A$		HE$			
A	AN$(5)		HAT$(3)		WAS$		THAT$				
B	BUT$(8)										
C											
D				HAD$							
E			HER$(7)				THE$		BE$		
F	FOR$(3)									OF$	
G											
H	HA$(3)				WHICH$						
I	IN$(1)		HIS$		WITH$		THIS$				
J											
K											
L											
M											
N	NOT$	IN$				AND$				ON$	
O	ON$(9)										FOR$
P											
Q											
R						ARE$		HER$		OR$	FROM$
S		IS$				AS$					
T	THA$(6)	IT$				AT$					
U									BUT$		
V				HAVE$							
W	WH$(4)										
X											
Y	YOU$								BY$		
Z											

Disadvantage of Tries

- Table-based tries typically waste large amounts of memory
- Often table-based tries are used for the first few layers, while lower levels use a less memory intensive data structure
- These days memory is less of a problem so table-based tries are acceptable for some applications
- There are many implementations of tries each suited to a particular task
- For DNA tries are not too wasteful (although maybe for proteins)

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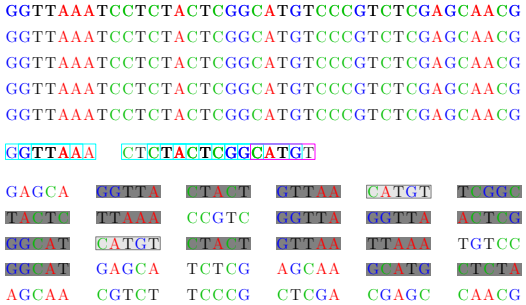
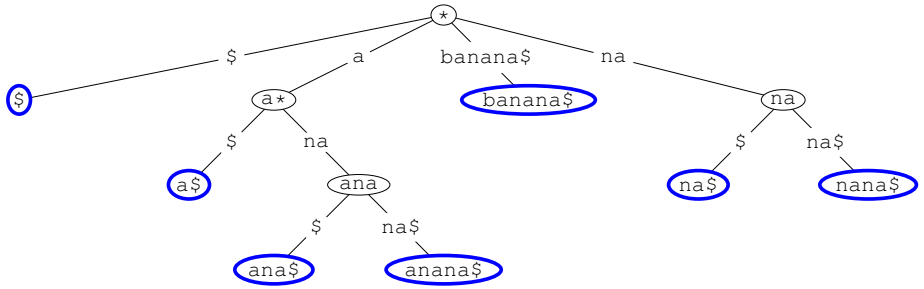


Suffix Tree

- Suffix tree is a trie of all suffixes of a string
- E.g. banana

{$, a$, na$, ana$, nana$, anana$, banana$}



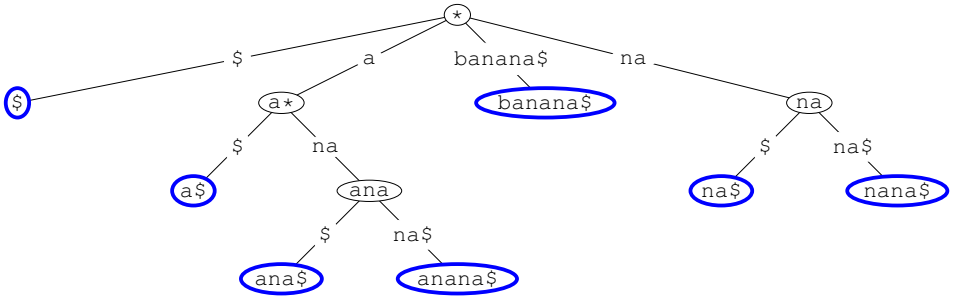
Importance of Suffix Tree

- Fast algorithms to compute Suffix trees (and the related suffix array) are relative new (15-20 year old) data structures
- They are very important for string-based algorithms
- The classic application is in finding a match for a query string, Q, in a text, T

String Matching

- To find a match of a query string, $Q = an$, in a text, $T = banana$, we can first construct the suffix tree of the string T we then simple look up the query, Q, using the trie

{$, a$, na$, ana$, nana$, anana$, banana$}



Complexity of Suffix Trees

- Using a regular trie for a suffix tree would typically use far too much memory to be useful
- However, by using pointers to the original text it is possible to build a suffix tree using $O(n)$ memory where n is the length of the text
- Furthermore (and rather incredibly) there is a linear time ($O(n)$) algorithm to construct the trie
- The algorithm is not however trivial to understand

Uses of Suffix Trees

- Suffix trees are efficient whenever it is likely that you will do multiple searches
- Exact word matching is in itself a very important application
- Suffix trees in combination with dynamic programming (which we look at next) can be used to do inexact matching (finding the match with the smallest edit distance)
- Suffix trees get used in bioinformatics, advanced machine learning algorithms, . . .
- In case you are interested there are also prefix trees containing all prefixes

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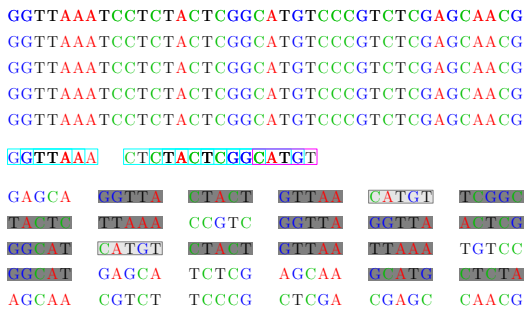


New Generation Sequencers

- The estimated cost of sequencing the human genome in 2005 was $\$10\,000\,000$
- To reduce the cost there was and is a drive to produce new sequencing machines
- These tend to read much shorter sections of DNA (e.g. 20-100nt)
- Can these be re-assembled?

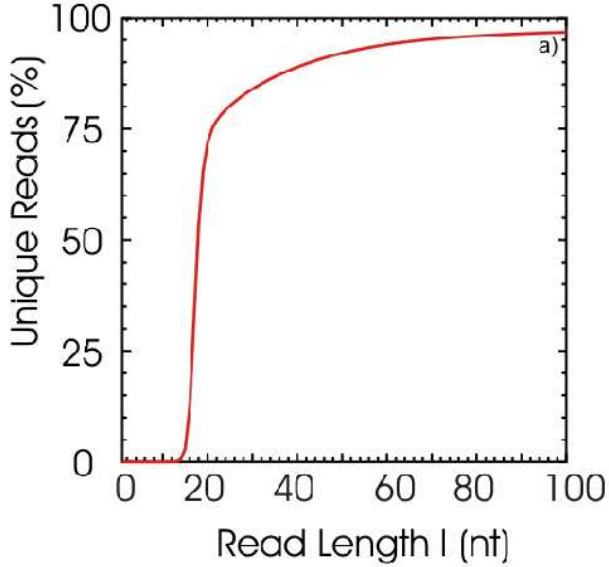
Repeats

- The difficulty of assembly is caused by repeats

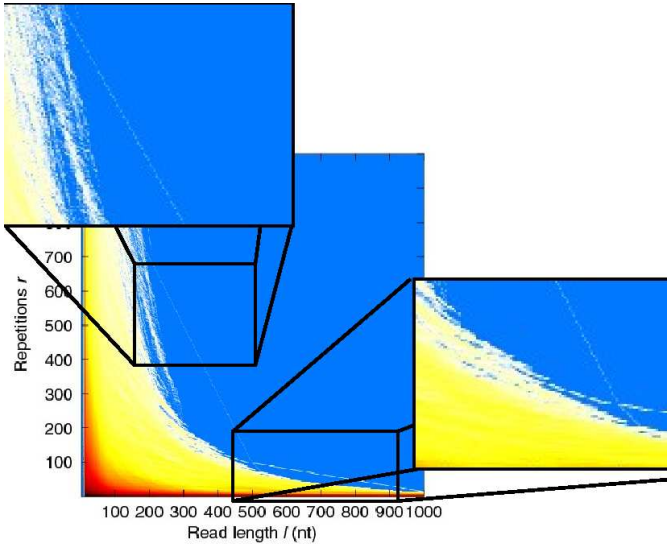
  ATACCACCATGCCTCCTTGCTCCAATCCACCATCAAGGCG

- How many repeats are there in the human genome? (Incidentally the human genome is 3.2 billion base **pairs**)
- You can read DNA in either directions meaning there are effectively 6.4 billion bases
- This is an important question for developing new sequencing technologies

Repeats in Human Genome



Repeats Structure



Computing Repeats

- A naive program would take n^2 operations where $n = 6.4 \times 10^9$
- If we used this we would still be waiting for the program to finish
- Could not answer this question a few years ago—not because computers weren't powerful, but because the algorithms had not been developed
- However using suffix arrays counting repeats was made easy

Summary

- The incredibly swift development in algorithms has leveraged the advances in computer power and smart algorithms
- Digital trees of tries, though simple can significantly speed up algorithms for re-assembly
- Suffix trees are conceptually simple (digital trees of all suffixes)
- They can vastly speed up operations such as matching
- What is complicated is the algorithm to construct them efficiently (though today they are widely available in programming languages such as python)